Year 11 Mathematics

Handling Data 4

4.3 Presenting Data

Presenting Data

STARTER · 5 MIN

Date: _____ **Title:** Presenting Data

LEARNING OBJECTIVES

- Draw and read bar charts and dual bar charts
- Draw and interpret pie charts
- Read and interpret line graphs

STARTER *Answer in your book. No calculators.*

1. Work out 360 ÷ 30
2. Work out 12 × 15
3. What fraction of 360° is 90°?
4. Work out $\frac{1}{3}$ of 360

Starter — Answers

STARTER

Tick or correct in green pen

1. **12**

2. **180**

3. $\frac{90}{360} = \frac{1}{4}$

4. $360 \div 3 = \mathbf{120}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can draw and read bar charts and dual bar charts	☐
SILVER	I can draw and interpret pie charts	☐
GOLD	I can read and interpret line graphs	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Presenting Data

INTRODUCTION · 3 MIN

Choose a chart that shows the data clearly. Always label axes and give a title or key.

Bar chart

Equal width bars with gaps.

Height of bar = frequency

Dual bar chart

Two bars side by side to compare.

Needs a key

Pie chart

Angle = frequency × 360 ÷ total.

All angles add to 360°

Line graph

Points joined to show change over time.

Shows the trend

Examples 1 and 2 — Bar Charts and Dual Bar Charts

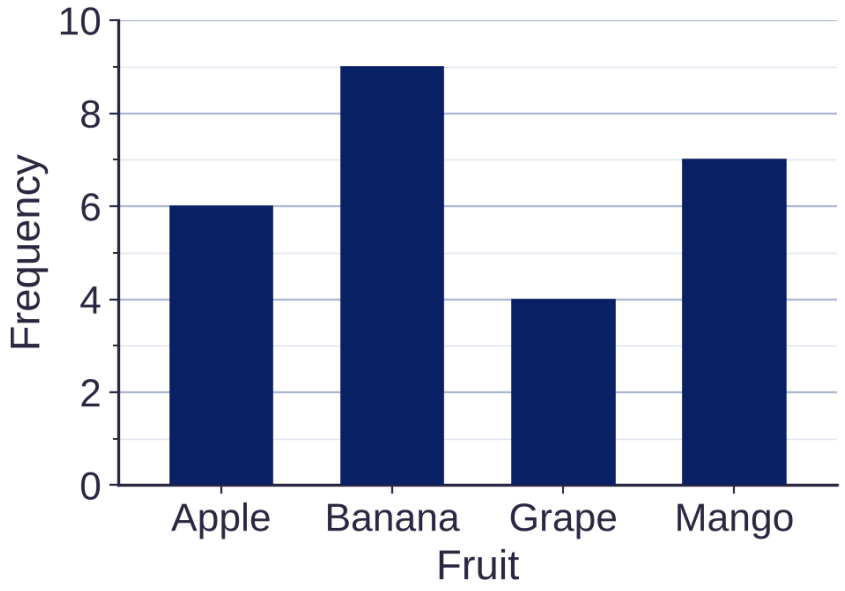
EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

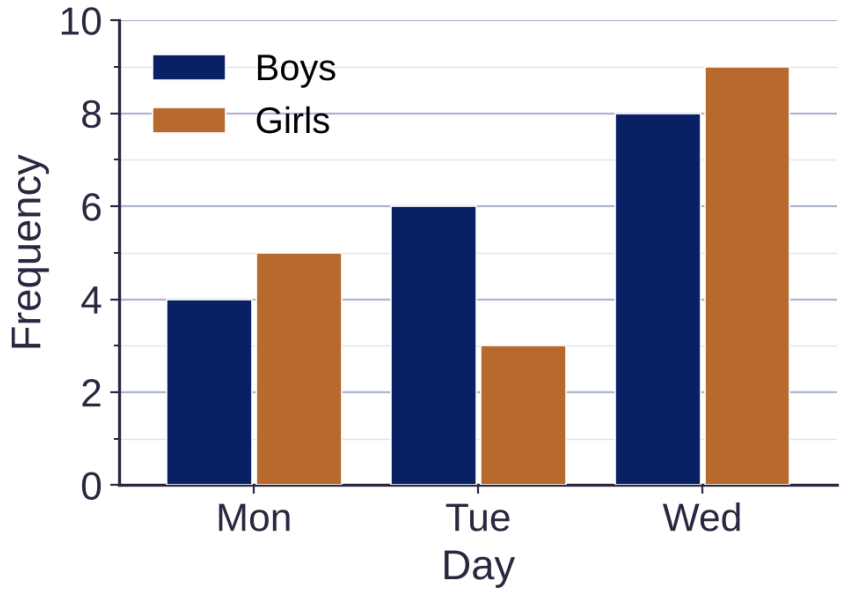
Example 1

How many students chose banana?



Example 2

How many students visited on Wednesday?



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Read across from the top of the bar

Banana **= 9**

Tip: use the gridlines to read across.

Example 2

Boys 8 and girls 9

8 + 9

17 students

Tip: use the key to tell the bars apart.

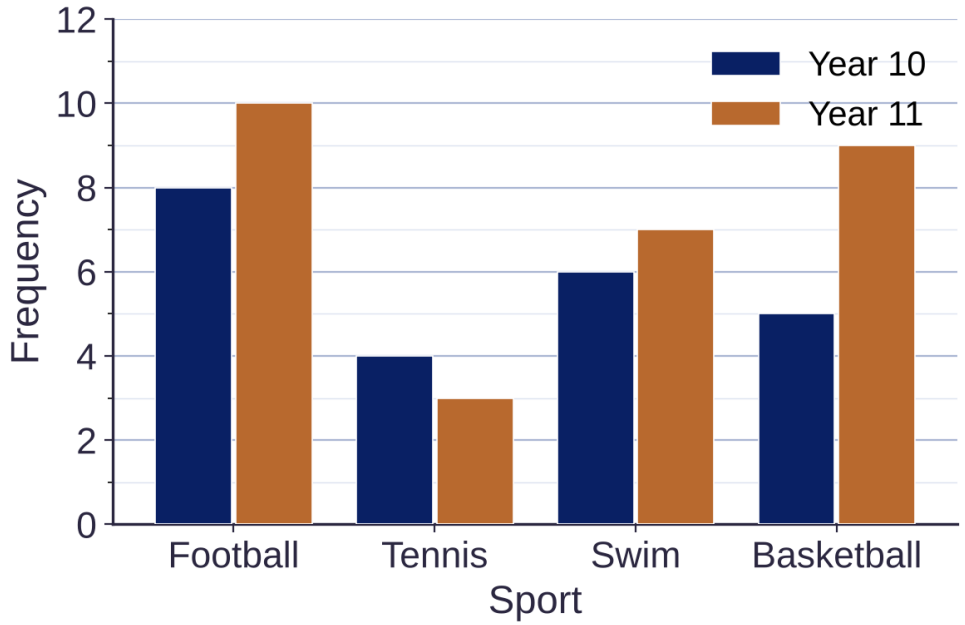
Task 1

PRACTICE · 9 MIN

BRONZE *Use the dual bar chart*

1. How many Year 10 chose football?
2. How many Year 11 chose basketball?
3. Which sport was most popular in Year 11?
4. How many chose swimming altogether?
5. How many more Year 11 than Year 10 chose basketball?

★ How many Year 10 students were asked?



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **8**

2. **9**

3. **Football**

4. 6 + 7 = **13**

5. 9 − 5 = **4**

★ 8 + 4 + 6 + 5 = **23**

Examples 3 and 4 — Pie Charts

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

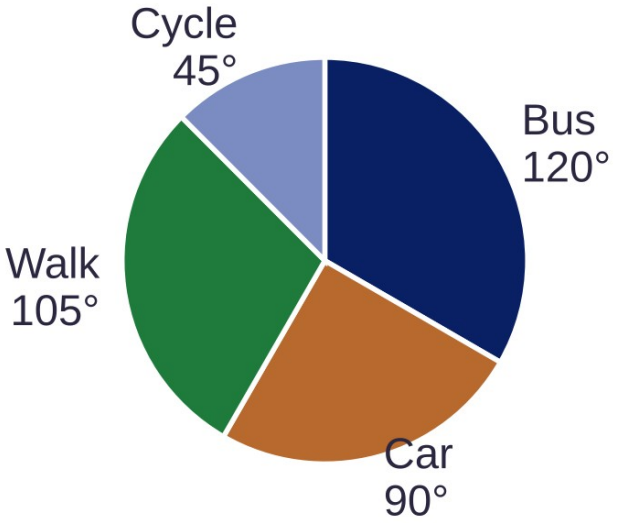
Hint $\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360°$

Example 3

60 students: Bus 20, Car 15, Walk 25. Work out the angles for a pie chart.

Example 4

72 people were asked. How many walk?



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

360 ÷ 60 = 6° each

Bus: 20 × 6 = 120°

Car: 15 × 6 = 90°

Walk: 25 × 6 = 150°

Tip: check the angles add to 360°.

Example 4

72 ÷ 360 = 0.2 people per degree

105 × 0.2

21 people

Tip: find the people per degree first.

Task 2

PRACTICE · 9 MIN

SILVER *Draw a pie chart for the table*

Hint $\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360°$

1. Find the total frequency
2. Work out the number of degrees per person
3. Work out the angle for tea
4. Work out the angle for coffee
5. Work out the angles for juice and water

Drink	Frequency
Tea	12
Coffee	18
Juice	6
Water	9

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $12 + 18 + 6 + 9 = \mathbf{45}$

2. $360 \div 45 = \mathbf{8°}$

3. $12 \times 8 = \mathbf{96°}$

4. $18 \times 8 = \mathbf{144°}$

5. $6 \times 8 = 48°$, $9 \times 8 = \mathbf{72°}$

Examples 5 and 6 — Line Graphs and Interpreting Charts

EXAMPLES · 5 MIN

GOLD

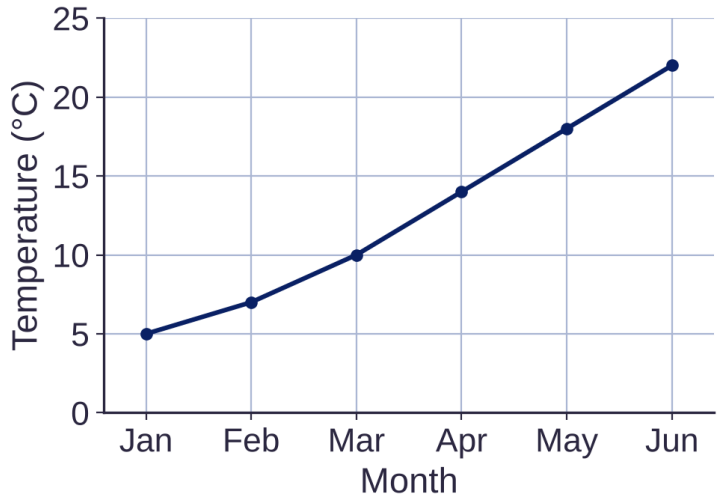
Exam-style: show all your working

Hint $\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360°$

Example 5

(2 marks)

Describe the trend. Find the increase from Jan to Jun.



Example 6

(2 marks)

A pie chart shows 120 students. Football has an angle of 135°. How many chose football?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

The temperature rises every month

22 − 5

Increase **= 17°C**

Tip: trend = what happens overall.

Example 6

135 ÷ 360 × 120

45 students

Tip: angle ÷ 360 gives the fraction.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360°$

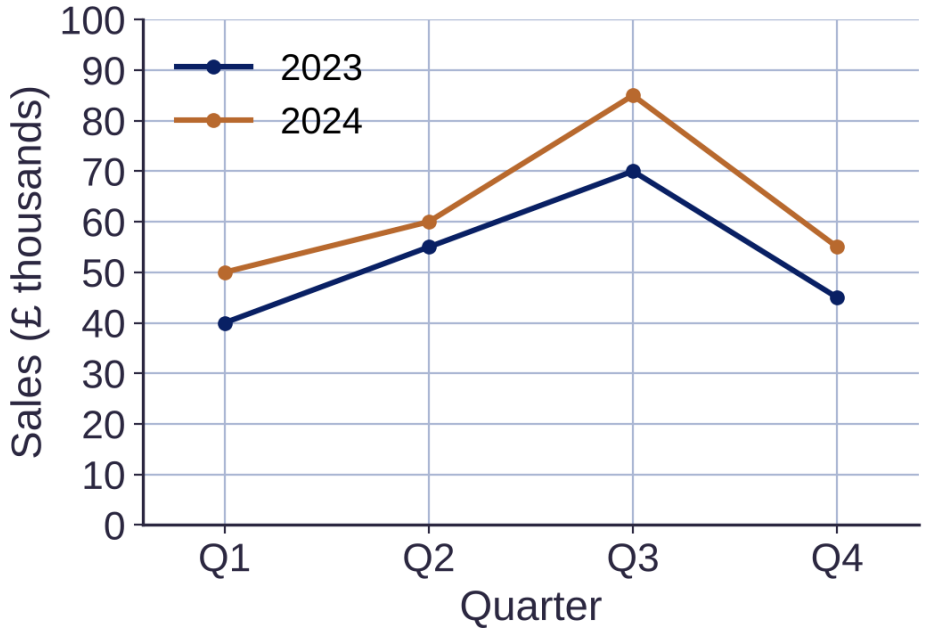
1. Write down the sales in Q3 2024. (1 mark)

2. Find the increase from Q1 to Q3 in 2023. (2 marks)

3. Compare the sales in the two years. (2 marks)

4. Find the total sales in 2024. (2 marks)

5. A pie chart shows 180 people. Walk is 80°. How many walk? (2 marks)



Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. **£85 000**

2. $70 - 40 =$ **£30 000**

3. 2024 higher every quarter, **both peak in Q3**

4. $50 + 60 + 85 + 55 =$ **£250 000**

5. $80 \div 360 \times 180 =$ **40**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Bars of height 4, 7 and 3. Find the total frequency.

SILVER Total frequency 30. Find the angle for a frequency of 5.

GOLD A pie chart shows 60 people. Car is 90°. How many?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $4 + 7 + 3 = \mathbf{14}$

SILVER $360 \div 30 = 12°, \quad 5 \times 12 = \mathbf{60°}$

GOLD $90 \div 360 \times 60 = \mathbf{15}$

Exam Practice — Bar Charts

EXAM PRACTICE · 10 MIN

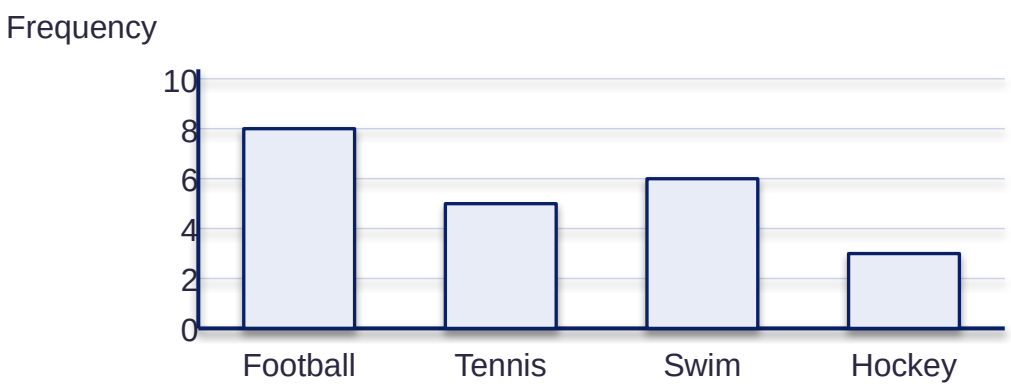
BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

The bar chart shows favourite sports.



(a) How many students altogether? (1)

.....

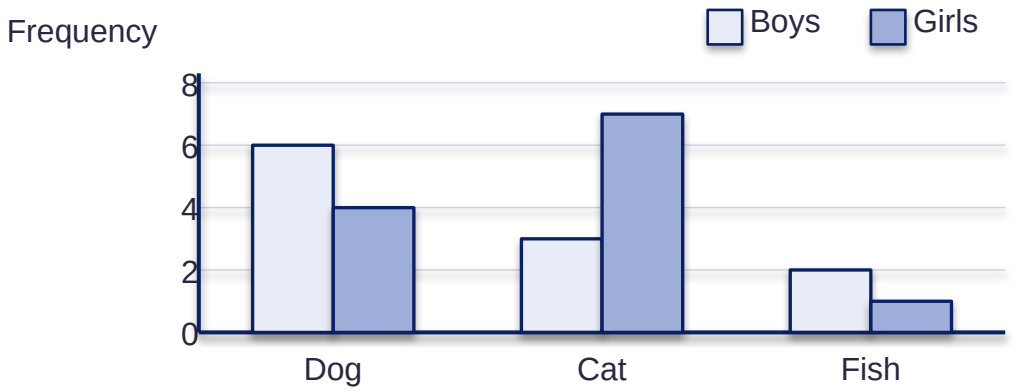
(b) Which sport is the mode? (1)

.....

Question 2

(2 marks)

Favourite pets of a class:



(a) How many girls chose cat? (1)

.....

(b) How many students chose dog? (1)

.....

Exam Practice — Pie Charts

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360°$

Question 3

(3 marks)

90 people were asked how they travel.

Work out the angle for each sector of a pie chart.

Travel	Frequency	Angle
Bus	30	
Car	25	
Walk	20	
Cycle	15	

Question 4

(2 marks)

A pie chart shows how 240 people voted.

The angle for Party A is 105°.

How many people voted for Party A?

.....

Exam Practice — Line Graphs and Comparing Pie Charts

EXAM PRACTICE · 10 MIN

GOLD

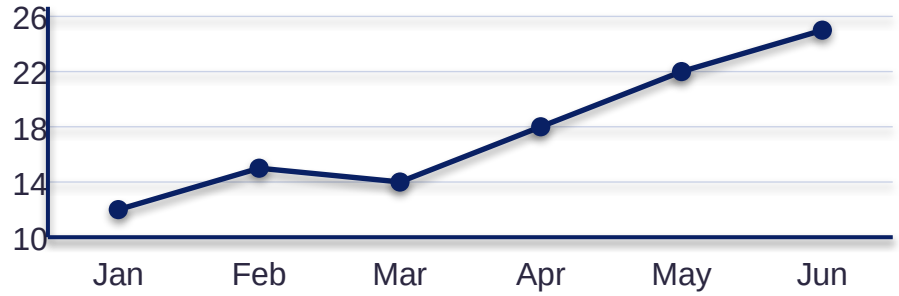
IGCSE exam-style: show all your working

Hint $\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360°$

Question 5

(3 marks)

Museum visitors (thousands):



(a) Describe the trend. (1)

.....

(b) Find the % increase from Jan to Jun, to 1 d.p. (2)

..... %

Question 6

(3 marks)

School A: 200 students. Tea is 90° of its pie chart.

School B: 360 students. Tea is 60° of its pie chart.

Sam says more students chose tea in School A.

Is Sam correct? Show your working.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 8 + 5 + 6 + 3 = 22 — B1

(b) football — B1

Tip: read across from the top of each bar.

Question 2

(a) 7 — B1

(b) 6 + 4 = 10 — B1

Tip: use the key to tell the bars apart.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

360 ÷ 90 = 4° per person — M1

Bus 120°, Car 100° — A1

Walk 80°, Cycle 60° — A1

Tip: check: the angles add up to 360°.

Question 4

$\frac{105}{360} \times 240$ — M1

= 70 — A1

Tip: find the fraction of 360°, then of 240.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) visitors increase over time — B1

(b) $25 - 12 = 13$ — M1

$\frac{13}{12} \times 100 = 108.3\%$ — A1

Tip: divide by the original amount.

Question 6

A: $\frac{90}{360} \times 200 = 50$ — M1

B: $\frac{60}{360} \times 360 = 60$ — M1

No: 60 > 50 — A1

Tip: a bigger angle does not mean more people.